Use Case: Checkers Home Menu

Iteration: 1

Primary Actor: Player

Goal In Context: Player is at the Home Menu screen for Checkers, where they can choose to enter the queue to play the game or view the leaderboard.

Preconditions:

- The player has logged in and is at the Player Profile Home Menu.
- The Checkers game is available.

Trigger: The player chooses to play Checkers and clicks on Checkers.

Scenario:

1. The player has logged into their account.
2. They are now at the Player Profile Home Menu.
3. They choose to play Checkers and click onto the game.
4. They are brought to the Home Menu screen for Checkers.
5. They can now choose to view the leaderboard or join a queue.

Post Conditions: The player is directed to view the leaderboard for Checkers, or to join the queue for the game.

Exceptions:

- **E1:** The Checkers module is unavailable, system shows an error message and the player is redirected to the Player Profile Home Menu.

Priority: High

When Available: After the player logs in and arrives at the Checker Home Menu.

Frequency of Use: Frequent, shown every time the player wants to play Checkers.

Channel to Actor: UI

Secondary actors: None

Channels to secondary actors: N/A

Open Issues:

- Should there be more options for the game at the Home Menu for the player to choose from?

Use Case: Checkers game queue

Iteration: 1

Primary Actor: Player

Goal In Context: The player would be able to join a queue to get matched up with an opponent.

Preconditions:

- The player has logged in and is at the Checkers Home Menu.
- The matchmaking service is available.

Trigger: The player chooses “Join Queue” from the Checkers Home Menu.

Scenario:

1. The player has logged into their account.
2. They choose to play Checkers and selects “Join Queue”.
3. The matchmaking system searches for an opponent.
4. If they are matched up with another player, they are directed to the Checkers game instance.

Post Conditions: The player is successfully matched up with another player, or they could not be matched up because no match is found.

Exceptions:

- **E1:** No other player wants to play Checkers at the same time and matching up fails.
- **E2:** The matchmaking system is down, the system shows an error message and redirects the player to the Checkers Home Menu.

Priority: High

When Available: When a player chooses to join a queue.

Frequency of Use: Moderate, depends on the player’s choices and wait times.

Channel to Actor: UI

Secondary actors: Matchmaking service

Channels to secondary actors: System-to-system communication

Open Issues:

- Should the player be able to choose to cancel queuing and get back to the game Home Menu?

Use Case: Game instance

Iteration: 1

Primary Actor: Player

Goal In Context: The player can play the Checkers game with an opponent.

Preconditions: The player has logged in and is matched up successfully with another player.

Trigger: The player chooses to play Checkers and another player is found.

Scenario:

1. The player clicks “Join Queue” and is matched up successfully with another player.
2. They are now at a screen where they can play Checkers and are notified that the game has started.
3. The players take turns and play the game until the game ends (win, lose, or draw).
4. After the game, they will be directed to the aftermath screen.
5. They can choose to return to the Checkers Home Menu or view the leaderboard.

Post Conditions: The game results are recorded and the aftermath screen is shown after the game is finished.

Exceptions:

- **E1:** The opponent disconnects, the system notifies the player, the game is ended.
- **E2:** The player’s connection is lost, the system stops the game.

Priority: High

When Available: After matchmaking.

Frequency of Use: Depends if the player can successfully match up with an opponent.

Channel to Actor: UI with real-time updates

Secondary actors: Opponent player

Channels to secondary actors: Network communication

Open Issues:

- Should the player be able to rematch with the opponent?

Use Case: No match found

Iteration: 1

Primary Actor: Player

Goal In Context: The player is notified that there is no match currently and they can retry or exit.

Preconditions: The player chooses to join a Checkers Game Queue.

Trigger: The player waits in the queue but the system could not find an opponent with in a certain time period.

Scenario:

1. The player chooses to join a Checkers Game Queue.
2. The matchmaking service searches for an opponent.
3. After a timeout period, the system still could not find an opponent.
4. The player is shown the “No Match Found” notification.
5. They have options to retry or exit back to the Checkers Home Menu.

Post Conditions: The player can try again or they are directed back to the Home Menu.

Exceptions:

- **E1:** There is an error while getting the player’s queue status, the system shows an error message and the player is redirected back to the Checkers Home Menu.

Priority: Medium

When Available: When there are low numbers of players.

Frequency of Use: Occasional, depends on the number of players.

Channel to Actor: UI notification

Secondary actors: None

Channels to secondary actors: N/A

Open Issues:

- What if there is no match for the player for a very long time and the player really wants to play?
- Should we include a function to have a game with AI?

Use Case: Checkers leaderboard

Iteration: 1

Primary Actor: Player

Goal In Context: The player would be able to see a screen with the leaderboard data for Checkers.

Preconditions:

- The player chooses Checkers from their profile Home Menu.
- The leaderboard functionality is available.

Trigger: The player is at the Checkers Home Menu and chooses “Leaderboard”.

Scenario:

1. The player choose Checkers on their Profile Home Menu.
2. They want to see the leaderboard and select “Leaderboard”.
3. They are shown with the leaderboard for Checkers with the following data:
    - **Rank:** The player’s ranking based on win rate or score.
    - **Username:** The display name of the player.
    - **Total Wins:** The total number of games the player has won.
    - **Win Rate:** The percentage of games won out of total games played.
    - **Longest Win Streak:** The highest number of consecutive wins of the player.
    - **Games Played:** The total number of games the player has played.
    - **Last Played Date:** The date when the player last played.
    - **Moves to Win (Average):** The average number of moves the player used to win a game.

Post Conditions: The player has now viewed the leaderboard for Checkers with their stats and also other players’ stats, they can now go back to the Checkers Home Menu.

Exceptions:

- **E1:** The system fails to retrieve leaderboard data, the player is shown an error message.
- **E2:** The leaderboard functionality is not available, the player is informed and the have the choice to come back later or return to the Checkers Home Menu.

Priority: Medium

When Available: When the player is at the Checkers Home Menu.

Frequency of Use: Occasional, depends on if the player wants to check the leaderboard.

Channel to Actor: UI display of the leaderboard

Secondary actors: Leaderboard database/service

Channels to secondary actors: System-to-database communication

Open Issues:

- Should there be a friends system?
- Should the player be able to sort or filter the leaderboard?